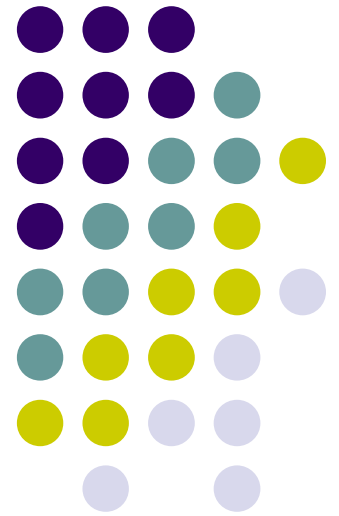
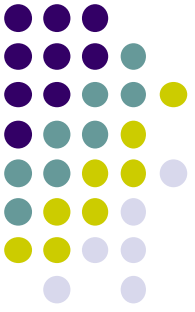


WIA1005 Network Technology Foundation

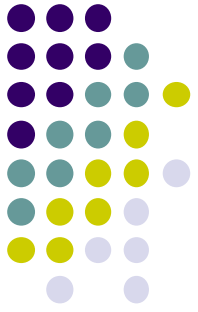
Chapter 13 Build and Secure Small Network





Contents

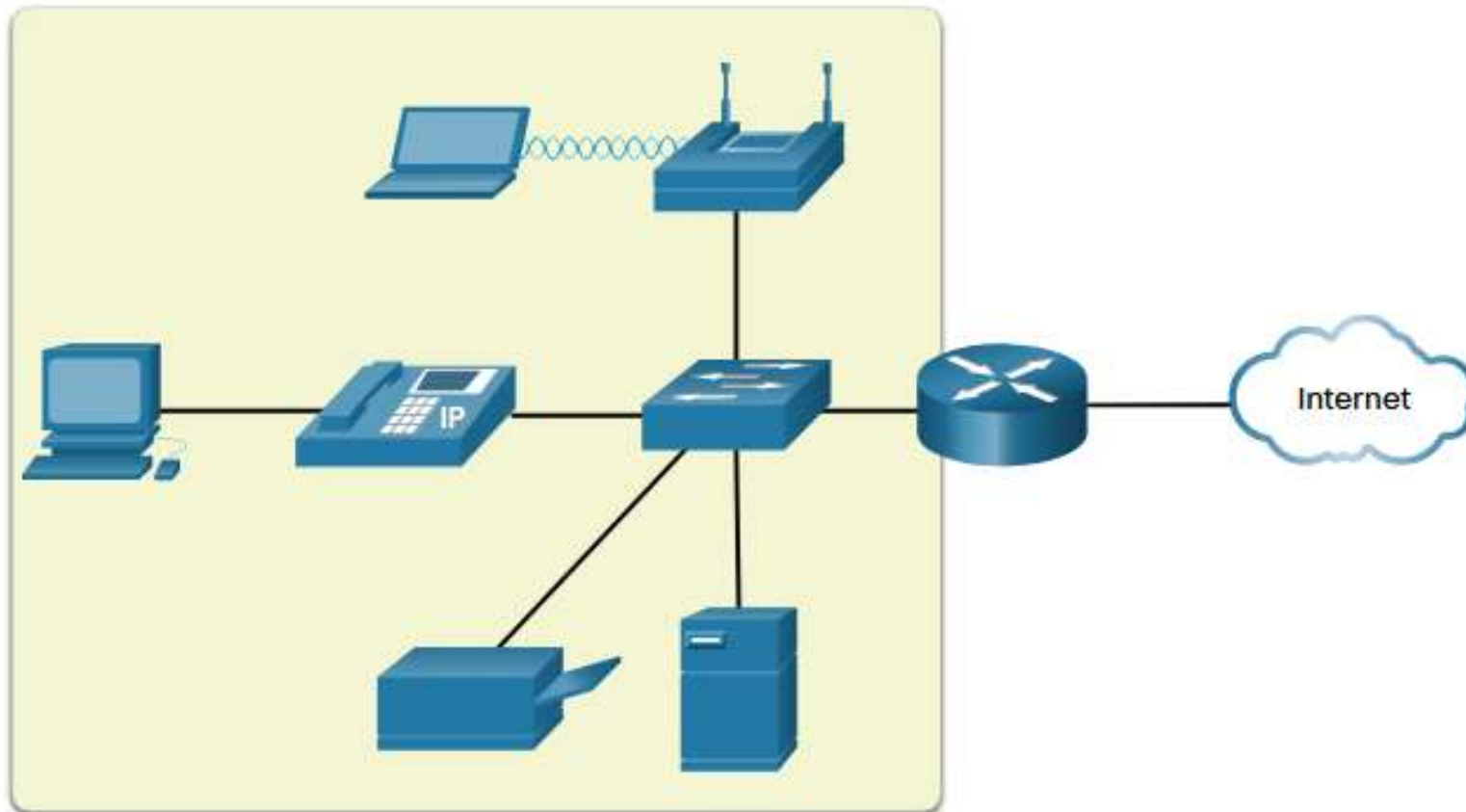
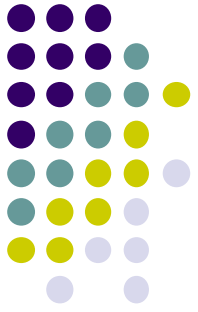
- Introduction
- Small Network Applications and Protocols
- Troubleshooting
- Security Threats
- Network Attacks
- Network Attack Mitigations

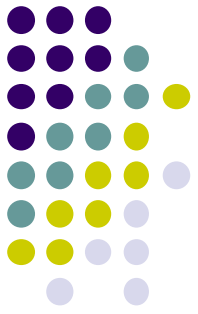


Introduction

- The design of a small network is usually simple. The number and type of devices included are significantly reduced compared to that of a larger network.
- The small network requires a router, a switch, and a wireless access point to connect wired and wireless users, an IP phone, a printer, and a server.
- Small networks typically have a single WAN connection provided by DSL, cable, or an Ethernet connection.
- Small networks are usually managed by a local IT technician.

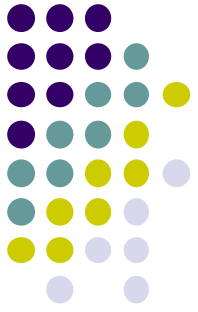
Introduction





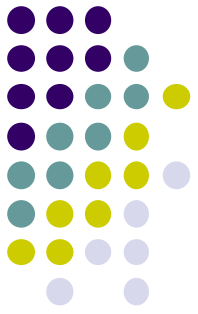
Introduction

- There are a number of factors that need to be considered when implementing a small network:
 - **Cost** - Cost of intermediary device is determined by its capacity and features. This includes the number and types of ports available and the backplane speed.
 - Other factors that influence the cost are network management capabilities, embedded security technologies, and optional advanced switching technologies.
 - The expense of cable runs required to connect every device on the network must also be considered.
 - Another key element affecting cost considerations is the amount of redundancy to incorporate into the network.



Introduction

- **Speed and Types of Ports/Interfaces** - Choosing the number and type of ports on a router or switch is a critical decision. Choosing Layer 2 devices that can accommodate increased speeds allows the network to evolve without replacing central devices.
- **Expandability**
 - Expandability – Fixed / Modular. Care must be taken to select the appropriate modules and interfaces for the specific media.
- **Operating System Features and Services**



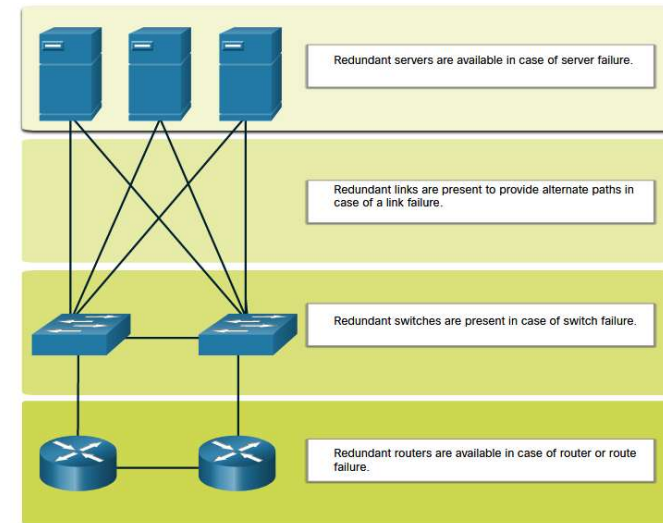
Introduction

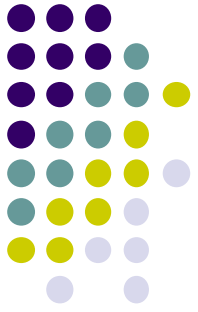
- When implementing a small network, it is recommended that you plan, document, and maintain an **IP addressing scheme** based on device type.
- The use of a planned IP addressing scheme makes it easier to identify a type of device and to troubleshoot problems.

Device Type	Assignable IP Address Range	Summarized as ...
Default gateway (Router)	192.168.x.1 - 192.168.x.2	192.168.x.0/30
Switches (max 2)	192.168.x.5 - 192.168.x.6	192.168.x.4/30
Access points (max 6)	192.168.x.9 - 192.168.x.14	192.168.x.8/29
Servers (max 6)	192.168.x.17 - 192.168.x.22	192.168.x.16/29
Printers (max 6)	192.168.x.25 - 192.168.x.30	192.168.x.24/29
IP Phones (max 6)	192.168.x.33 - 192.168.x.38	192.168.x.32/29
Wired devices (max 62)	192.168.x.65 - 192.168.x.126	192.168.x.64/26
Wireless devices (max 62)	192.168.x.193 - 192.168.x.254	192.168.x.192/26

Introduction

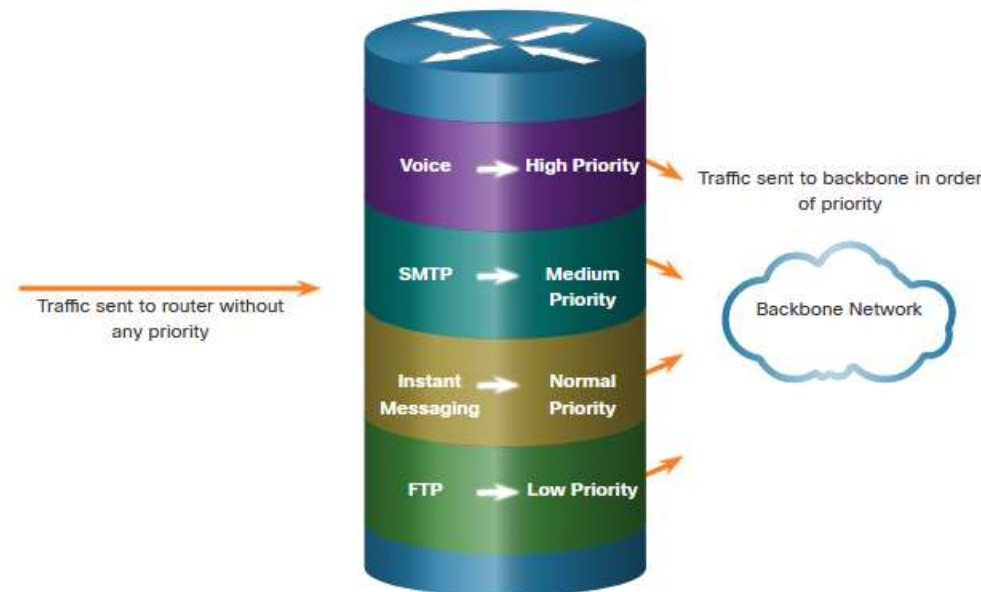
- Another important part of network design is **reliability**. Even small businesses often rely heavily on their network for business operation. A failure of the network can be very costly.
- In order to maintain a high degree of reliability, **redundancy** is required in the network design.
- Redundancy can be accomplished by installing duplicate equipment, but it can also be accomplished by supplying duplicate network links for critical areas.

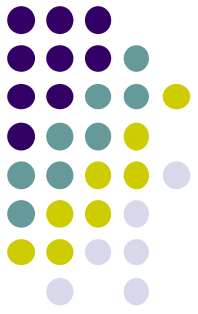




Introduction

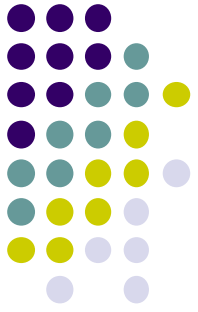
- The goal for a good network design, even for a small network, is to enhance the productivity of the employees and minimize network downtime.
- The network administrator should consider the various types of traffic and their treatment in the network design.





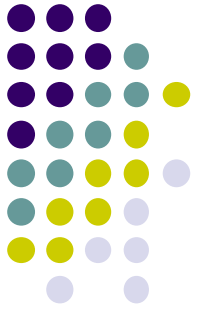
Small Network Applications and Protocols

- Network administrators commonly require access to network devices and servers. The two most common remote access solutions are Telnet and Secure Shell (SSH).
- SSH is used to establish a secure remote access connection between an SSH client and other SSH-enabled devices:
 - **Network device** - The network device must support SSH to provide remote access SSH server services to clients.
 - **Server** - The server must support remote access SSH server services to clients.



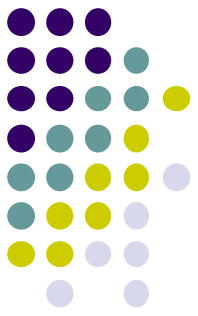
Small Network Applications and Protocols

- Businesses today are increasingly using IP telephony and streaming media to communicate with customers and business partners.
- Many organizations are enabling their employees to work remotely.
- The network administrator must ensure the proper equipment is installed in the network and that the network devices are configured to ensure priority delivery.
- Besides, network administrator needs to plan for the growth of the small network.



Small Network Applications and Protocols

- To scale a network, several elements are required:
 - **Network documentation** - Physical and logical topology
 - **Device inventory** - List of devices that use or comprise the network
 - **Budget** - Itemized IT budget, including fiscal year equipment purchasing budget
 - **Traffic analysis** - Protocols, applications, and services and their respective traffic requirements should be documented



Small Network Applications and Protocols

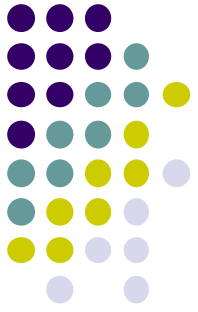
- As the network grows, it becomes important to determine how to manage network traffic. It is important to understand the type of traffic that is crossing the network as well as the current traffic flow.

Wireshark · Protocol Hierarchy Statistics - wireshark_9E2D6026-50E6-4476-ADE1-7A92AAAA61FB_20190815100157_a19512

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	39501	100.0	24008911	184 k	0	0	0
Ethernet	100.0	39501	2.3	553014	4238	0	0	0
Internet Protocol Version 6	0.0	18	0.0	720	5	0	0	0
Internet Protocol Version 4	97.7	38593	3.2	771860	5916	0	0	0
User Datagram Protocol	47.1	18587	0.6	148696	1139	0	0	0
Simple Service Discovery Protocol	0.6	253	0.3	67080	514	253	67080	514
Session Traversal Utilities for NAT	2.8	1115	0.2	55808	427	1115	55808	427
QUIC (Quick UDP Internet Connections)	42.8	16902	51.2	12295632	94 k	6933	2458683	18 k
NetBIOS Name Service	0.0	3	0.0	150	1	3	150	1
NetBIOS Datagram Service	0.0	8	0.0	1736	13	0	0	0
Domain Name System	0.7	285	0.1	17874	137	285	17874	137
Data	0.0	18	0.0	11808	90	18	11808	90
Bootstrap Protocol	0.0	3	0.0	900	6	3	900	6
Transmission Control Protocol	50.5	19950	41.8	10043477	76 k	11348	4813481	36 k
Secure Sockets Layer	20.2	7975	41.3	9916053	76 k	7301	7978903	61 k
Malformed Packet	0.3	125	0.0	0	0	125	0	0
Hypertext Transfer Protocol	0.0	12	0.0	5408	41	4	1441	11
Data	2.9	1164	0.3	69637	533	1164	69637	533
Internet Control Message Protocol	0.1	56	0.0	2240	17	56	2240	17
Address Resolution Protocol	2.3	890	0.1	24920	191	890	24920	191

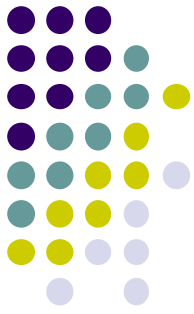
No display filter.

Close Copy Help



Small Network Applications and Protocols

- To determine traffic flow patterns, it is important to:
 - Capture traffic during peak utilization times to get a good representation of the different traffic types.
 - Perform the capture on different network segments and devices as some traffic will be local to a particular segment.
- A network administrator must be aware of how network use is changing. Documenting snapshots for employees in a small network over a period of time is very useful to identify evolving protocol requirements and associated traffic flows. A shift in resource utilization may require the network administrator to adjust network resource allocations accordingly.

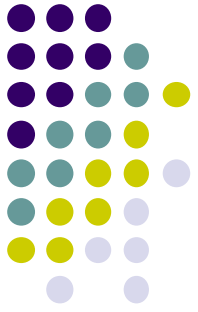


Troubleshooting

- Troubleshooting is used to analyse the problem, determine the cause of the error, implement a plan of action, and resolve the network issue.

Table 1: Six Steps of the Troubleshooting Methodology

Step	Title	Description
1	Identify the Problem	The first step in the troubleshooting process is to identify the problem. While tools can be used in this step, a conversation with the user is often very helpful.
2	Establish a Theory of Probable Causes	After you have talked to the user and identified the problem, you can try and establish a theory of probable causes. This step often yields more than a few probable causes to the problem.
3	Test the Theory to Determine Cause	Based on the probable causes, test your theories to determine which one is the cause of the problem. A technician will often apply a quick procedure to test and see if it solves the problem. If a quick procedure does not correct the problem, you might need to research the problem further to establish the exact cause.
4	Establish a Plan of Action to Resolve the Problem and Implement the Solution	After you have determined the exact cause of the problem, establish a plan of action to resolve the problem and implement the solution.
5	Verify Full System Functionality and Implement Preventive Measures	After you have corrected the problem, verify full functionality and, if applicable, implement preventive measures.
6	Document Findings, Actions, and Outcomes	In the final step of the troubleshooting process, document your findings, actions, and outcomes. This is very important for future reference.

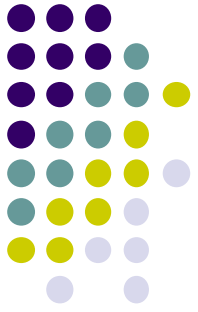


Troubleshooting

- When a problem has been solved and a solution implemented, it is important to verify the system operation. **ping**, **tracert** and **show** are some of the verification commands
- **ping** command is an effective way to test connectivity. The ping command uses the Internet Control Message Protocol (ICMP) and verifies Layer 3 connectivity.

```
C:\> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time=0ms TTL=128
Reply from 127.0.0.1: bytes=32 time=4ms TTL=128
Reply from 127.0.0.1: bytes=32 time=3ms TTL=128
Reply from 127.0.0.1: bytes=32 time=3ms TTL=128

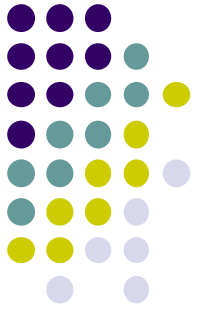
Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms
```

Troubleshooting

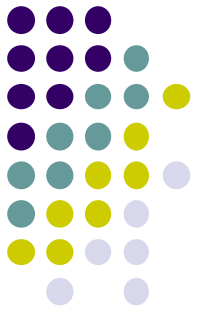
- Traceroute is useful in troubleshooting routing loops and determining the exact next-hop router.
- An **ICMP "time exceeded"** indicates that a router in the path has seen and discarded the packet. An **ICMP "destination unreachable"** indicates that a router has received the packet, but discarded it because it could not be delivered.

```
C:\> tracert
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name
Options:
    -d                Do not resolve addresses to hostnames.
    -h maximum_hops   Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                Force using IPv4.
    -6                Force using IPv6.
C:\>
```



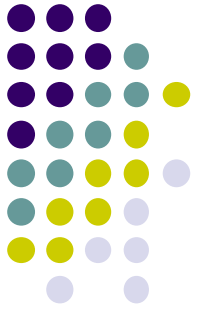
Troubleshooting

- One of the most effective tools for monitoring and troubleshooting network performance is to establish a **network baseline**.
- Creating an effective network performance baseline is accomplished over a period of time.
- Measuring performance at varying times and loads will assist in creating a better picture of overall network performance.



Troubleshooting

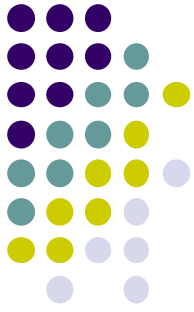
- The **show** commands display relevant information about the configuration and operation of the device.
 - show running-config
 - show interfaces / show ip interface
 - show arp
 - show ip route
 - show protocols
 - show version
- The **ipconfig** or **ipconfig /all** commands are used to view IP address. (ifconfig - linux, ifconfig en0 - mac)
- The **arp -a** command lists all devices in ARP cache.



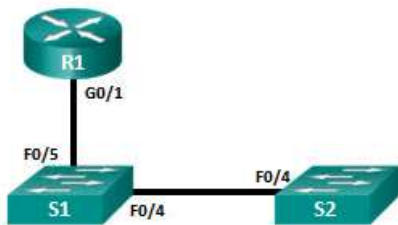
Troubleshooting

- Cisco Discovery Protocol (CDP) is a proprietary Layer 2 protocol that is used to gather information about Cisco devices which share the same network.
- The device sends periodic CDP advertisements to connected devices. These advertisements share information about the **type of device** that is discovered, the **name** of the devices, and the **number and type** of the interfaces.
- CDP can also be used as a network discovery tool to assist in network design decisions, troubleshooting, and making changes to equipment.

Troubleshooting



- **show cdp neighbors** command provides:
 - Device ID
 - Port ID
 - Device (Switch/Router)
 - Platform



```
S1# show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

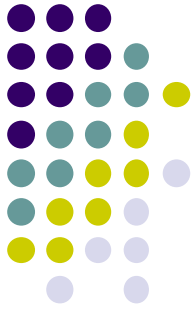
Device ID         Local Intrfce   Holdtme    Capability  Platform  Port ID
S2                Fas 0/4        158        S I         WS-C2960- Fas 0/4
R1                Fas 0/5        136        R B S I     CISCO1941 Gig 0/1
```

```
S2# show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability  Platform  Port ID
S1                Fas 0/4        173        S I         WS-C2960- Fas 0/4
```

Troubleshooting



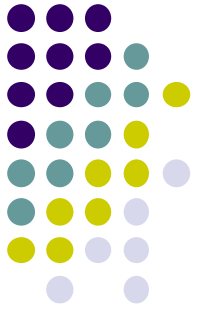
```
R1# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability  Platform  Port ID
S1                 Gig 0/1        122        S I         WS-C2960- Fas 0/5

R1# show cdp neighbors detail
-----
Device ID: S1
Entry address(es):
  IP address: 192.168.1.2
Platform: cisco WS-C2960-24TT-L, Capabilities: Switch IGMP
Interface: GigabitEthernet0/1, Port ID (outgoing port): FastEthernet0/5
Holdtime : 136 sec

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE7,
RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2014 by Cisco Systems, Inc.
Compiled Thu 23-Oct-14 14:49 by prod_rel_team

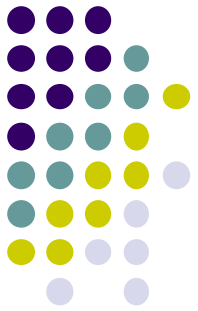
advertisement version: 2
Protocol Hello: OUI=0x00000C, Protocol ID=0x0112; payload len=27,
value=00000000FFFFFFFFF010221FF0000000000000002291210380FF0000
VTP Management Domain: ''
Native VLAN: 1
```



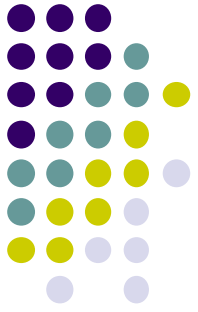
Troubleshooting

- The IOS **debug** command provides valuable information when troubleshooting or verifying system operations.
- All debug commands are entered in privileged EXEC mode. The filter command is used to include only the relevant information.
- Commands such as debug all or debug ip packet a generate a substantial amount of output.
- Disable the debugging process when enough data has been captured:
 - no debug
 - undebug
 - undebug all

Troubleshooting

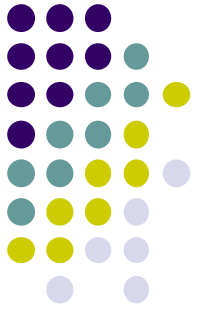


```
R1# debug ip icmp
ICMP packet debugging is on
R1# ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
*Nov 13 12:56:08.147: ICMP: echo reply rcvd, src 10.0.0.10, dst 10.0.0.1,
topology BASE, dscp 0 topoid 0
*Nov 13 12:56:08.151: ICMP: echo reply rcvd, src 10.0.0.10, dst 10.0.0.1,
topology BASE, dscp 0 topoid 0
*Nov 13 12:56:08.151: ICMP: echo reply rcvd, src 10.0.0.10, dst 10.0.0.1,
topology BASE, dscp 0 topoid 0
*Nov 13 12:56:08.151: ICMP: echo reply rcvd, src 10.0.0.10, dst 10.0.0.1,
topology BASE, dscp 0 topoid 0
*Nov 13 12:56:08.151: ICMP: echo reply rcvd, src 10.0.0.10, dst 10.0.0.1,
topology BASE, dscp 0 topoid 0
R1# undebug all
All possible debugging has been turned off
R1#
```

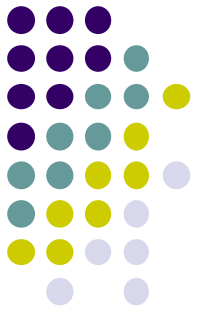
Troubleshooting

- Administrative can access to a router remotely on virtual console (VTY) lines using SSH or Telnet. However, log messages are not displayed by default on virtual console lines.
- To display log messages on a terminal (virtual console)
 - **terminal monitor**
- To stop logging messages on a terminal
 - **terminal no monitor**



Security Threats

- **Information theft** is breaking into a computer to obtain confidential information. Information can be used or sold for various purposes.
- **Data loss and manipulation** is breaking into a computer to destroy or alter data records
- **Identity theft** is a form of information theft where personal information is stolen for the purpose of taking over someone's identity.
- **Disruption of service** is preventing legitimate users from accessing services to which they are entitled.



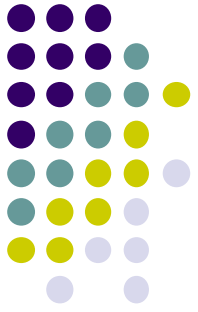
Security Threats

- Vulnerability is the degree of weakness in a network or a device.
- There are three primary vulnerabilities or weaknesses: technological, configuration, and security policy.

Technological Vulnerabilities

Vulnerability	Description
TCP/IP Protocol Weakness	• Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), and Internet Control Message Protocol (ICMP) are inherently insecure. • Simple Network Management Protocol (SNMP) and Simple Mail Transfer Protocol (SMTP) are related to the inherently insecure structure upon which TCP was designed.
Operating System Weakness	• Each operating system has security problems what must be addressed. • UNIX, Linux, Mac OS, Mac OS X, Windows Server 2012, Windows 7, Windows 8 • They are documented in the Computer Emergency Response Team (CERT) archives at http://www.cert.org
Network Equipment Weakness	Various types of network equipment, such as routers, firewalls, and switches have security weaknesses that must be recognized and protected against. Their weaknesses include password protection, lack of authentication, routing protocols, and firewall holes.

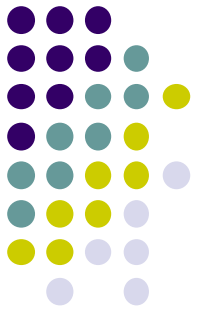
Security Threats



Configuration Vulnerabilities

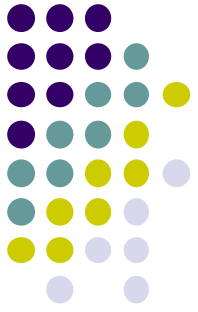
Vulnerability	Description
Unsecured user accounts	User account information may be transmitted insecurely across the network, exposing usernames and passwords to threat actors.
System accounts with easily guessed passwords	This common problem is the result of poorly created user passwords.
Misconfigured internet services	Turning on JavaScript in web browsers enables attacks by way of JavaScript controlled by threat actors when accessing untrusted sites. Other potential sources of weaknesses include misconfigured terminal services, FTP, or web servers (e.g., Microsoft Internet Information Services (IIS), and Apache HTTP Server.
Unsecured default settings within products	Many products have default settings that create or enable holes in security.
Misconfigured network equipment	Misconfigurations of the equipment itself can cause significant security problems. For example, misconfigured access lists, routing protocols, or SNMP community strings can create or enable holes in security.

Security Threats



Policy Vulnerabilities

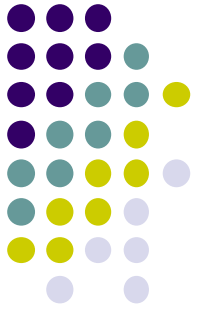
Vulnerability	Description
Lack of written security policy	A security policy cannot be consistently applied or enforced if it is not written down.
Politics	Political battles and turf wars can make it difficult to implement a consistent security policy.
Lack of authentication continuity	Poorly chosen, easily cracked, or default passwords can allow unauthorized access to the network.
Logical access controls not applied	Inadequate monitoring and auditing allow attacks and unauthorized use to continue, wasting company resources. This could result in legal action or termination against IT technicians, IT management, or even company leadership that allows these unsafe conditions to persist.
Software and hardware installation and changes do not follow policy	Unauthorized changes to the network topology or installation of unapproved application create or enable holes in security.
Disaster recovery plan is nonexistent	The lack of a disaster recovery plan allows chaos, panic, and confusion to occur when a natural disaster occurs or a threat actor attacks the enterprise.



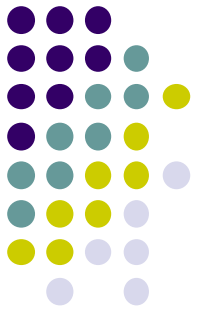
Security Threats

- An equally important vulnerable area of the network to consider is the physical security of devices.
- The four classes of physical threats are as follows:
 - **Hardware threats** - This includes physical damage to servers, routers, switches, cabling plant, and workstations.
 - **Environmental threats** - This includes temperature extremes (too hot or too cold) or humidity extremes (too wet or too dry).

Security Threats

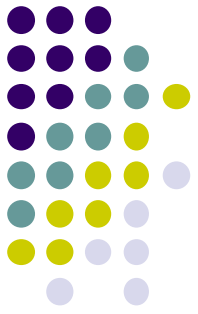


- **Electrical threats** - This includes voltage spikes, insufficient supply voltage (brownouts), unconditioned power (noise), and total power loss.
- **Maintenance threats** - This includes poor handling of key electrical components (electrostatic discharge), lack of critical spare parts, poor cabling, and poor labeling.



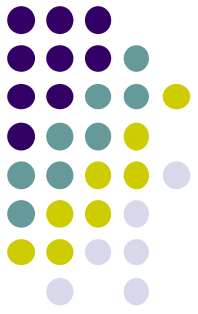
Network Attacks

- Malware is short for malicious software. It is code or software specifically designed to damage, disrupt, steal, or inflict “bad” or illegitimate action on data, hosts, or networks.
- Viruses, worms, and Trojan horses are types of malware.
- **Viruses** - a type of malware that propagates by inserting a copy of itself into, and becoming part of, another program. It spreads from one computer to another, leaving infections as it travels.
- **Worms** – standalone software that replicate functional copies of themselves automatically



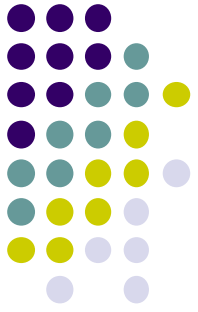
Network Attacks

- **Trojan Horses** - It is a harmful piece of software that looks legitimate. Users are typically tricked into loading and executing it on their systems. After it is activated, it can achieve any number of attacks on the host, from irritating the user to damaging the host. It can create back doors to give malicious users access to the system. Trojan horses must spread through user interaction such as opening an email attachment or downloading and running a file from the internet.



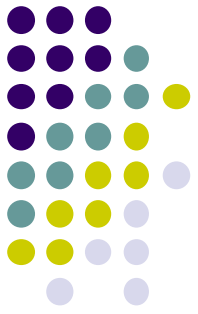
Network Attacks

- **Reconnaissance attacks** - The discovery and mapping of systems, services, or vulnerabilities. A threat actor can use internet tools, such as the **nslookup** and **whois** utilities, to easily determine the IP address. Then, a threat actor may use a ping sweep tool, such as **fping** or **gping**.
- **Access attacks** exploit known vulnerabilities in authentication services, FTP services, and web services to gain entry to web accounts, confidential databases, and other sensitive information. An access attack allows individuals to gain unauthorized access to information that they have no right to view. Access attacks can be classified into four types: password attacks, trust exploitation, port redirection, and man-in-the middle.



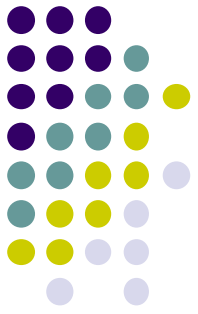
Network Attacks

- **Password Attacks**
 - Brute-force attacks
 - Trojan horse attacks
 - Packet sniffers
- **Trust Exploitation** - a threat actor uses unauthorized privileges to gain access to a system
- **Port Redirection** - a threat actor uses a compromised system as a base for attacks against other targets.
- **Man-in-the-Middle** - a threat actor is positioned in between two legitimate entities in order to read or modify the data that passes between the two parties.



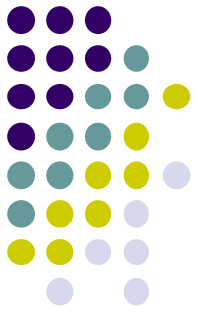
Network Attacks

- **Denial of service (DoS) attacks** are the most publicized form of attack and among the most difficult to eliminate. DoS attacks prevent authorized people from using a service by consuming system resources.
- A DDoS is similar to a DoS attack, but it originates from multiple, coordinated sources. For example, a threat actor builds a network of infected hosts, known as zombies. A network of zombies is called a **botnet**.
- The threat actor uses a command and control (CnC) program to instruct the botnet of zombies to carry out a DDoS attack.



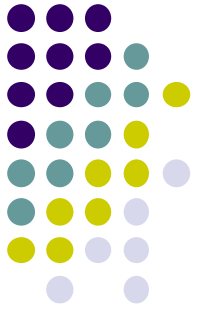
Network Attack Mitigations

- To mitigate network attacks, you must first secure devices including routers, switches, servers, and hosts.
- Backing up device configurations and data is one of the most effective ways of protecting against data loss. A data backup stores a copy of the information on a computer to removable backup media that can be kept in a safe place.
- Keeping up to date with the latest developments can lead to a more effective defense against network attacks.
- **Authentication, Authorization, and Accounting (AAA)** network security services provide the primary framework to set up access control on network devices.



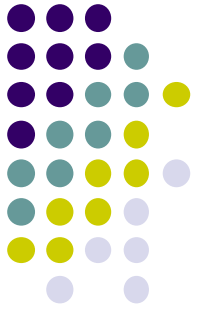
Network Attack Mitigations

Consideration	Description
Frequency	• Perform backups on a regular basis as identified in the security policy. • Full backups can be time-consuming, therefore perform monthly or weekly backups with frequent partial backups of changed files.
Storage	• Always validate backups to ensure the integrity of the data and validate the file restoration procedures.
Security	• Backups should be transported to an approved offsite storage location on a daily, weekly, or monthly rotation, as required by the security policy.
Validation	• Backups should be protected using strong passwords. The password is required to restore the data.



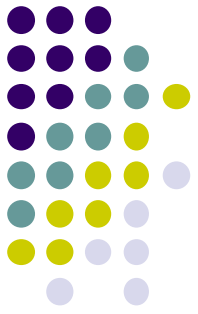
Network Attack Mitigations

- **VPN** - A router is used to provide secure VPN services with corporate sites and remote access support for remote users using secure encrypted tunnels.
- **ASA Firewall** - Provides stateful firewall services. It ensures that internal traffic can go out and come back, but external traffic cannot initiate connections to inside hosts.
- **IPS** - An intrusion prevention system (IPS) monitors incoming and outgoing traffic looking for malware, network attack signatures, and more. If it recognizes a threat, it can immediately stop it.



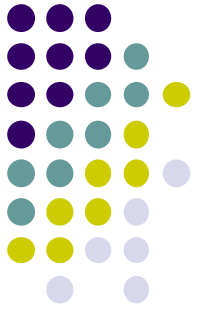
Network Attack Mitigations

- **ESA/WSA** - The email security appliance (ESA) filters spam and suspicious emails. The web security appliance (WSA) filters known and suspicious internet malware sites.
- **AAA Server** - This server contains a secure database of who is authorized to access and manage network devices.



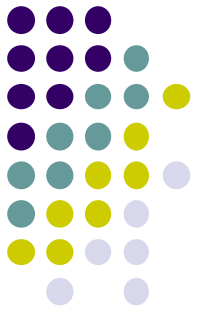
Network Attack Mitigations

- Firewall come packaged in various forms as below:
 - **Packet filtering** - Prevents or allows access based on IP or MAC addresses
 - **Application filtering** - Prevents or allows access by specific application types based on port numbers
 - **URL filtering** - Prevents or allows access to websites based on specific URLs or keywords
 - **Stateful packet inspection (SPI)** - Incoming packets must be legitimate responses to requests from internal hosts. Unsolicited packets are blocked unless permitted specifically. Include the capability to recognize and filter out specific types of attacks, such as denial of service (DoS)



Network Attack Mitigations

- **Endpoint Security** - Securing endpoint devices is one of the most challenging jobs of a network administrator because it involves human nature. A company must have well-documented policies in place and employees must be aware of these rules.
- Policies often include the use of antivirus software and host intrusion prevention. More comprehensive endpoint security solutions rely on network access control.



Network Attack Mitigations

```
Router(config)# service password-encryption
Router(config)# security password min-length 8
Router(config)# login block-for 120 attempts 3 within 60
Router(config)# line vty 0 4
Router(config-line)# password cisco
Router(config-line)# exec-timeout 5 30
Router(config-line)# transport input ssh
Router(config-line)# end
```

```
Router# show control-plane host open-ports
Active internet connections (servers and established)

```

Prot	Local Address	Foreign Address	Service	State
tcp	*:23	*:0	Telnet	LISTEN
tcp	*:80	*:0	HTTP CORE	LISTEN
udp	*:67	*:0	DHCPD Receive	LISTEN

```
Router# configure terminal
Router(config)# no ip http server
```